

Cardiometabolic Phenotypes and Predicted Trouble Sleeping in US Adults

SIADS 696

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Introduction:

Sleep is an essential physiological process that is necessary for physical restoration, mental wellbeing, and quality of life. Sleep problems can be associated with adverse physical, mental, and cardiovascular health outcomes. While previous machine learning studies have identified important individual risk factors for sleep-related problems, it still remains unclear whether patterns such as broader cardiometabolic profiles can provide additional predictive information beyond traditional demographic and clinical risk factors. This project seeks to identify distinct cardiometabolic phenotypes within the U.S. adult population (ages 18+) and evaluate whether phenotype membership improves prediction of self-reported sleep trouble beyond individual clinical markers alone.

Metabolic and sleep processes such as circadian rhythms are connected across neural, molecular, and cellular levels (Huang et al., 2011) suggesting that disruptions in metabolic regulation may influence sleep-wake regulation and overall sleep health. Metabolic Syndrome (MetS) is a clinically recognized cluster of interconnected factors that increase the risk of cardiovascular diseases and type 2 diabetes (Kassi et al., 2011). While some individual components of MetS have been studied in relation to sleep outcomes, less is known about whether distinct combinations of these metabolic factors alongside cardiovascular factors define population phenotypes with differing levels of sleep-related risk. Identifying such cardiometabolic phenotypes could improve prediction of sleep trouble and provide some clinically meaningful understanding of how cardiometabolic health influences sleep.

This study combines supervised and unsupervised machine learning approaches using data from the National Health and Nutrition Examination Survey (NHANES).

The supervised learning component of this project uses a subset of NHANES variables identified as clinically relevant to sleep trouble through literature review as features for input into Random Forest and XGBoost models. Model performance is evaluated using AUROC, sensitivity, specificity, F1 score, and confusion matrices. Feature importance analysis is used to identify the most influential predictors.

The unsupervised learning methods used in this project are k-means and agglomerative hierarchical clustering. Both clustering techniques are applied to cardiometabolic biomarkers including cardiovascular and some MetS criteria. Clustering is used to identify naturally occurring cardiometabolic phenotypes within the U.S. adult population. Variables are standardized prior to clustering and the optimal number of clusters is determined for hierarchical and k-means clustering using both elbow and silhouette methods. Initialization and data stability are tested to determine which clustering technique should be used to create the final phenotypes. These phenotype labels are then introduced as engineered features in the supervised model to test whether the phenotype membership improves prediction of sleep trouble beyond the initial supervised model.

Unsupervised methods found four stable, yet overlapping, clusters that were then added to the supervised learning model as the *phenotype* feature. These clusters were found to have high data stability and initialization stability but clusters were overlapping with likely many subjects on the boundary of the clusters. The supervised findings were that adding *phenotype* to the supervised model resulted in little to no improvement in model performance.

Related Work:

Huang and Huang (2023) used NHANES data to predict physician-identified sleep disorder status using machine learning models, including XGBoost, Random Forest, AdaBoost, and neural networks. Their study found that depression symptoms, weight, age, and waist circumference were among the strongest predictors of sleep disorder risk.

Our project builds on this work but shifts the focus from identifying individual predictors to testing whether broader cardiometabolic phenotype membership adds predictive value. Instead of only ranking variables, we first use unsupervised clustering to identify population subgroups and then evaluate whether those subgroup labels improve supervised sleep-trouble prediction.

Smiley, King, and Bidulescu (2019) examined the association between sleep duration and metabolic syndrome using NHANES 2013/2014 data. Their work supports the biological and epidemiological link between sleep patterns and metabolic risk.

Sabbagh et al. (2025) studied social jetlag, sleep, and metabolic syndrome in NHANES 2017-2020 adults. This study is relevant because it connects sleep timing and metabolic dysfunction in a nationally representative adult dataset. We instead use the same broad NHANES setting but we ask a different predictive question: whether cardiometabolic-lifestyle profiles can help classify sleep trouble. Rather than focusing on social jetlag as the main exposure, we focus on phenotype membership as an engineered feature for supervised learning.

Data Source(s):

The dataset used in this project is the National Health and Nutrition Examination Survey (NHANES), specifically the 2017-March 2020 pre-pandemic public-use files from the CDC/NCHS NHANES data portal. NHANES was selected because it combines questionnaire responses, physical examination measures, laboratory biomarkers, and demographic variables for a large sample of U.S. adults.

The raw NHANES files were downloaded as SAS transport files (.XPT) and loaded into Python as pandas DataFrames. The analysis used multiple NHANES modules, including demographics, sleep questionnaire data, physical activity questionnaire data, body measures, blood pressure, and laboratory biomarkers.

Important variables included the sleep-trouble outcome, age, sex, body mass index, waist circumference, physical activity indicators, sedentary minutes, glycohemoglobin, HDL cholesterol, total cholesterol, high-sensitivity C-reactive protein, uric acid, ALT, GGT, average systolic blood pressure, and average diastolic blood pressure. NHANES survey design variables, including MEC (Mobile Examination Center) weights, PSU, and strata, were retained in the cleaned dataset but were not used as machine learning predictors due to the importance of the interpretation of the weighting post-analysis.

Feature Engineering:

NHANES modules were merged using participant ID, and selected variables were renamed into readable feature names. The sample was filtered to adults aged 18 and older, invalid survey responses were handled as missing values, and rows with missing values in the final selected features were removed for complete-case modeling.

For supervised learning, the target variable was based on the NHANES sleep questionnaire item indicating whether a doctor or health professional had told the participant they had trouble sleeping or a sleep disorder. Predictors included demographic, lifestyle, body composition, cardiovascular, inflammatory, liver-enzyme, and metabolic variables.

For unsupervised learning, the sleep outcome was excluded, and cardiometabolic-lifestyle variables were standardized before clustering. Average systolic and diastolic blood pressure were engineered from available blood pressure readings. Fasting glucose, insulin, and triglycerides were excluded from the main feature set because they were only available for the fasting subsample.

After filtering to adults aged 18 and older, we created two main modeling dataframes: `metS_df` and `cardio_met`. The `metS_df` dataframe focused on metabolic syndrome-related variables, including waist circumference, blood pressure, HDL cholesterol, fasting glucose, and triglycerides. Because fasting glucose and triglycerides were only available for the NHANES fasting subsample, `metS_df` was smaller and used mainly as a sensitivity dataset. The `cardio_met` dataframe became the main dataset because it used broader MEC-based cardiometabolic and lifestyle features available for more participants. After complete-case filtering, `cardio_met` contained 7,207 adults, with 7,204 adults available for supervised sleep-trouble prediction.

Final features included age, sex, BMI, waist circumference, vigorous work activity, moderate work activity, walking or bicycling for transportation, sedentary minutes, glycohemoglobin, HDL cholesterol, total cholesterol, hs-CRP, uric acid, ALT, GGT, average systolic blood pressure, and average diastolic blood pressure. Phenotype membership from k-means clustering was later added as an engineered feature for the combined supervised model.

Supervised Learning:

Motivation

The goal of this project is to understand whether cardiometabolic health patterns can aid in self-reported sleep trouble in U.S. adults' predictions. Particularly, we want to investigate whether the cardiometabolic phenotypes derived in the unsupervised section add predictive value beyond the individual clinical and lifestyle markers. Our approach to determining this was done in three layers. The first was to build a baseline model that predicts sleep trouble from individual cardiometabolic, demographic, and lifestyle variables. In our next layer, we built a second model that adds the k-means phenotype label as an engineered feature and compares the two. The supervised task therefore gives us a more direct way to evaluate whether the patterns found in the data are related to sleep disorder status. If phenotype membership carries information that the individual variables do not, the supervised learning with phenotype model should outperform the baseline. If it doesn't then the signal is already contained in the NHANES variables.

Model Selection

Our modeling workflow was designed to answer three questions.

1. Is the predictive pattern mostly linear, or does it rely on more complex interactions among the features?
2. Which variables are actually carrying predictive information?
3. Does the unsupervised phenotype label add anything beyond the raw variables that created it?

Table 1: Dataset Imbalance Comparison

The selection of models derived from Smiley, King, and Bidulescu (2019) study, where we further developed their findings by training a supervised learning model based on the phenotypes found in the unsupervised learning model. We selected Logistic Regression, Random Forest, and XGBoost. We used Logistic Regression as an interpretable

SLQ050	Original (%)	Dropped Missing Values (%)
0	73.12	71.24
1	26.88	28.76

baseline to evaluate whether most of the predictive capabilities could be captured with a linear probabilistic approach. Random Forest was selected as its a bagged tree model that can deal with nonlinearities and interactions. XGBoost was included as boosted trees often do well on structured tabular data when there is meaningful nonlinear but also some noise.

Utilizing the variable selection established in the feature engineering portion, we ran our supervised model with ‘Sleep Disorder’ as our target variable. SLQ050, or the sleep disorder target variable is a binary variable where 1 indicates sleep disorder and 0 indicates no sleep disorder. We examined and found there was a moderate imbalance in our dataset, and ensured no missing values. We then analyzed how dropping missing values would affect class imbalance and found that the deletion did not strongly bias the outcome. Because the imbalance was moderate rather than severe, we did not apply synthetic oversampling. Categorical and numerical variables were then preprocessed using StandardScaler to ensure our models were not influenced by differences in feature scale and for ease of feature comparison after standardization. All baseline predictors were already numeric after cleaning so the preprocessing pipeline standardized them. When we later added Phenotype, we treated it as categorical and one-hot encoded it to ensure that the model would not assume an implied ordering among phenotype labels.

After running each model, we optimized the hyperparameter through GridSearchCV with 5-fold-cross-validation to give each model a fair chance under the same evaluation framework. We used F1 score instead of accuracy since accuracy can be misleading in an imbalanced dataset and our desired class was the minority in the imbalance dataset. An 80/20 train/test split was utilized, and a final analysis

Model Evaluation and Sensitivity Analysis

An evaluation of each supervised learning model was conducted mainly using the 5-fold cross-validation means and standardization. In addition to the aforementioned F1 metrics, precision, recall, balanced accuracy, and AUROC was utilized as each metric shows us the full story behind our data.

1. F1 - balances precision and recall into a single measure of overall performance on the positive class
2. Precision - how often positive predictions are right
3. Recall - how any sleep disorder cases are found
4. Balanced Accuracy - corrects for the class imbalance by weighting both class equally
5. AUROC - how well the model ranks cases overall before any threshold is chosen

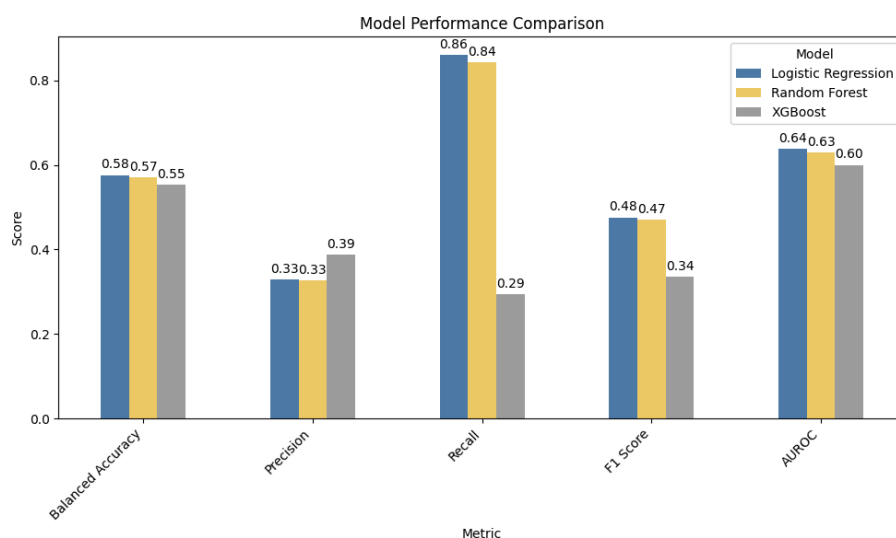


Figure 1: Supervised Learning Model Comparison

The 5-fold cross-validation results showed that Logistic Regression was the best performing supervised model overall, with the highest F1 score (0.476), recall (0.620), balanced accuracy (0.611), and AUROC (0.651), while also showing relatively low standard deviation across folds. Random Forest performed similarly but slightly worse on most metrics, whereas XGBoost performed substantially worse, particularly in recall and F1. Given these findings, we believe that the relationship between the selected predictors and sleep disorder may be captured reasonably well by a simpler linear model, rather than requiring a more complex tree-based approach.

This means it was more effective at detecting the positive class, which is important given that sleep disorder is the outcome of interest. However given the model's low precision score, it incorrectly classified many participants without sleep disorder as having sleep disorder. The model can be used for identifying possible risk patterns, but it should not be interpreted as a highly accurate diagnostic tool.

We utilized a confusion matrix for Logistic Regression to reflect the precision/recall pattern. Most true sleep-trouble cases are correctly identified (high recall), but a large number of true negatives are incorrectly flagged as positive (the source of the low precision). XGBoost shows the opposite tendency, correctly classifying many negatives while missing most positives. (Exact cell counts to be filled from the notebook output.)

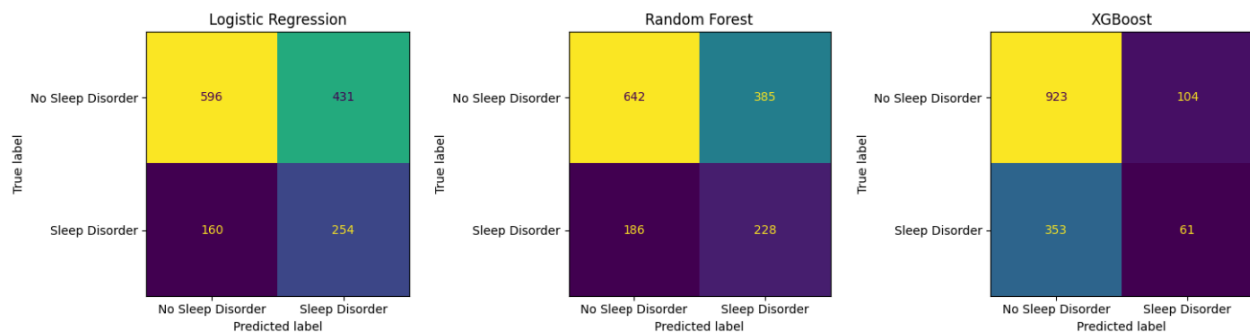


Figure 2: Confusion Matrices across all models

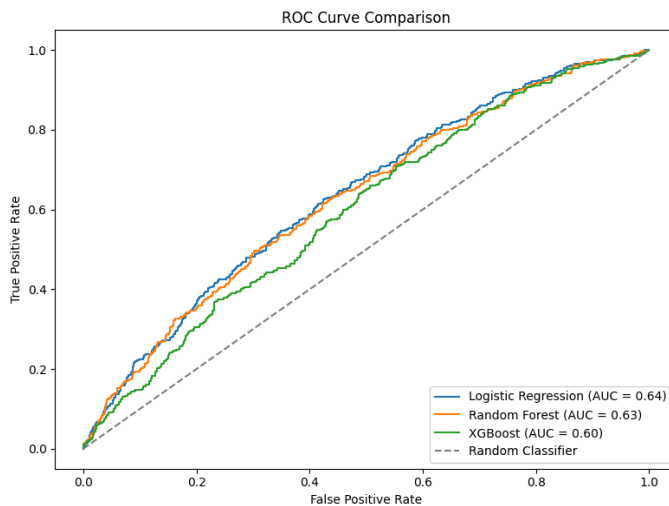


Figure 3: ROC across all models

We also evaluated the tuned models on the 20% held-out test set. As precision/recall trade off against each other and false negatives are costly (missing a sleep disorder case), we used a decision threshold of 0.40 rather than the default 0.50.

Feature Importance and Ablation Analysis

The most influential predictors, ranked by absolute coefficient, were: Waist Circumference, Age, Sex, Sedentary Minutes, and GGT. Positive-coefficient features push the prediction toward sleep trouble and negative-coefficient features push away from it. This may indicate that the model leaned most on body composition (waist circumference), demographics (age, sex),

sedentary behavior, and a metabolic/liver marker (GGT). This is consistent with the broader literature linking sleep problems to cardiometabolic health, essentially stating that the model is not relying on any single variable but on a mix of demographic, behavioral, and biomarker information.

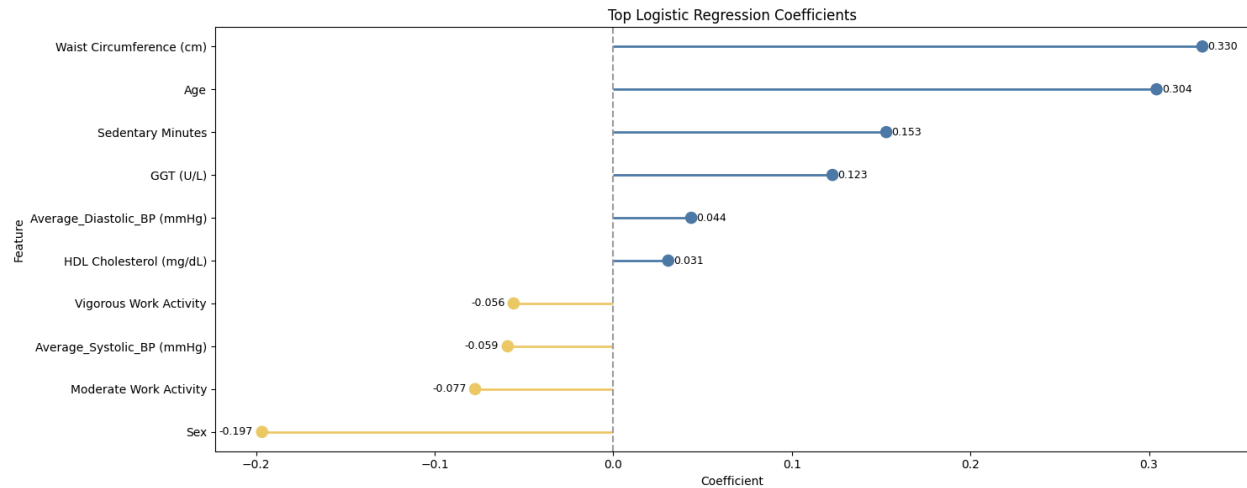


Figure 4: Feature importance in best model - Logistic Regression coefficients

We found that the predictors strongly shifted the model toward or away from a sleep disorder prediction. That interpretation becomes stronger once we look at ablation. We tested how much each group of features contributed, we removed feature groups from the Logistic Regression model and recorded 5-fold cross-validated metrics for each ablated configuration. It was found that the absolute F1 values in this analysis are lower than the main tuned model because the ablation routine used a simpler untuned Logistic Regression setup. Two findings stood out:

1. Removing the activity features caused the largest drop in F1 ($0.129 \rightarrow 0.090$), suggesting sedentary and work-activity variables contribute meaningfully to the model's ability to separate the classes.
2. Removing the top three predictors (Waist Circumference, Age, Sex) caused the largest drop in AUROC (0.651 to 0.619), confirming those variables carry the most ranking power. Removing biomarker features also reduced performance modestly.
3. The model relies on a combination of demographic, activity, and biomarker information.

The phenotype analysis is one of the most important findings in the whole supervised section. When we added the Phenotype label as an extra predictor, the results barely moved. Logistic Regression with phenotype produced the same test set metrics as the baseline Logistic Regression model: balanced accuracy (0.576), precision (0.329), recall (0.860), F1 (0.476), AUROC (0.638). Random Forest improved only slightly in F1, from 0.470 to 0.472, and XGBoost improved from 0.335 to 0.353, but neither changed the overall ranking.

This is significant as the phenotype label was descriptively useful but predictively ineffective. While it helped determine patterns in the population, the original predictors were already in the model's phenotype label and it added almost no new information. The unsupervised clusters seem to be a repackaging signal that is already present in age, waist circumference, sedentary behavior, and biomarkers rather than contributing a new axis of predictive information.

Threshold Analysis (Sensitivity Analysis and Model Tradeoffs)

Our sensitivity analysis and model tradeoff was determined by a threshold sensitivity analysis. We performed the analysis on the best model and examined the decision threshold from 0.10 to 0.90. A clear precision and recall trade-off was found after examining the precision, recall, F1, and balanced accuracy at each value. At low thresholds the model achieves near perfect recall but very low precision (it labels almost everyone positive). As the threshold rises, precision improves while recall falls sharply. We ultimately chose 0.40 as it gives the best F1 and prioritizes recall. It should be noted that if the model is being used like a screening tool, then missing sleep disorder cases are

costly, and a lower threshold such as 0.40 makes sense because it keeps recall high. If the model is being used in a setting where false alarms create a larger problem, then a threshold closer to 0.50 becomes more attractive.

Failure Analysis

We examined specific test records of the final Logistic Regression model misclassified that were taken from its highest confidence false positives and false negatives. As we examined our failure analysis we found that the model is learning a real pattern, but it is not always learning the right one. In particular, it tends to equate general cardiometabolic risk with sleep disorder risk. It struggles with participants who report a sleep disorder but do not look metabolically high risk. We identified three distinct categories of failure.

Failure 1: Systematic over-prediction from a high-risk cardiometabolic profile (false positive). We saw cases where the participant matched almost every pattern the model associates with the positive class and it directly reflected the linear decision boundary that pushed the prediction high. We suspect that the model has learned to equate a strong cardiometabolic risk profile with sleep trouble, and combined with the low threshold it over predicts the positive class.

Case 1687: Participant's true label was no sleep disorder but the model predicted sleep disorder with high probability

Probable reason: Model learned to equate strong cardiometabolic risk with sleep disorder risk, even when the true label was negative

Future improvement: Add more sleep-specific variables so that the model does not rely so heavily on general metabolic risk

Table 2: Failure Case 1

Failure 2: Missing a true case that looks healthy (false negative). For example, when a participant looks like a standard negative on the variables the model uses most. This may indicate that the model has no strong feature signal for young, lean, metabolically healthy individuals who report sleep trouble. This could be a case where true positive falls outside the risk pattern that the model learned.

Case 4334: Participant's true label was sleep disorder, but the model predicted no sleep disorder with low probability

Probable reason: Record looked relatively low risk on the variables the model relied on most, including younger age, low BMI, low waist circumference, and other healthy biomarkers. The model had little signal for participants whose sleep disorder was not in the cardiometabolic profile it learned.

Future improvement: Include features such as sleep duration, stress, mental health, or medication

Table 3: Failure Case 2

Failure 3: Outlier biomarker values dominating the linear prediction (false positive). The unusually large biomarker values, after standardization, produced a large positive linear combination. Logistic Regression is sensitive to extreme standardized inputs which may have affected an anomalous record with several outlying biomarkers.

Case 2437: Participant's true label was no sleep disorder, but the model predicted sleep disorder with probability around 0.80

Probable reason: record contained unusually large biomarker and blood pressure values

Future improvement: Further inspect extreme values, consider more robust preprocessing, and evaluate whether nonlinear models or feature transformations handle these edge cases better.

Table 3: Failure Case 3

Supervised Learning Model with Phenotype Findings

We re-ran the full supervised pipeline after adding the k-means phenotype label as an extra engineered feature, giving 18 predictors over the same 7,204 participants, and compared it to the baseline.

Model	Feature Set	Threshold	Balanced Accuracy	Precision	Recall	F1 Score	AUROC
Logistic Regression	Baseline	0.4	0.576	0.329	0.860	0.476	0.638
Random Forest	Baseline	0.4	0.570	0.326	0.843	0.470	0.630
XGBoost	Baseline	0.4	0.553	0.387	0.295	0.335	0.600
Logistic Regression + Phenotype	Baseline + Phenotype	0.4	0.576	0.329	0.860	0.476	0.638
Random Forest + Phenotype	Baseline + Phenotype	0.4	0.572	0.327	0.848	0.472	0.626
XGBoost + Phenotype	Baseline + Phenotype	0.4	0.561	0.395	0.319	0.353	0.600

Table 4: Final Supervised Model Evaluation Comparison

The results show that adding the phenotype label produced little to no improvement. For the best-performing model, Logistic Regression, all test metrics remained exactly the same after phenotype was added. Our key finding is that phenotypes helped us describe and understand broad participant groupings, but they did not provide substantial new predictive information once the underlying cardiometabolic and behavioral variables were already included directly in the model. *In short, the phenotype feature did not significantly improve the supervised model.* Although the unsupervised clustering produced interpretable cardiometabolic groups, those group labels did not add much predictive information once the original variables such as age, BMI, waist circumference, blood pressure, inflammation markers, and activity measures were already included. This tells us that the supervised model was already capturing most of the useful signal from the individual cardiometabolic variables, compressing those variables into phenotype labels did not strengthen the model's prediction.

Unsupervised Methods:

We used two unsupervised clustering approaches to identify cardiometabolic phenotypes, K-means clustering (centroid-based) and Agglomerative Hierarchical clustering (tree-based). Both methods are first evaluated across a range of k values to determine the best k using silhouette score and elbow plots then evaluated for data stability given the optimal k. The phenotype labels are then compared across methods to evaluate the similarity between

phenotypes produced. Based on the data stability and clustering quality k-means was then selected as the final model and used to produce cluster summaries and cluster visualizations.

These two clustering options were chosen because of the importance of interpretability in the resulting phenotypes and because a single cluster assignment was needed. The single assignment is important since the phenotype label is used later in for model comparison. K-means was chosen because of its simplicity and interpretability, it assigns each subject to one cluster and will be beneficial for creating clinically interpretable phenotypes. Agglomerative hierarchical clustering was chosen as a non-centroid based approach that builds a grouping structure that focuses on minimizing within-cluster variance. We also decided to use hierarchical clustering because it is very interpretable and a dendrogram can be used to see which groups are merged together and how far apart or close together groups are. These two methods then allowed us to see whether or not similar groupings are produced under different methods and then select which method to move forward with in order to create our final phenotype groupings.

The features used in this unsupervised stage include age, BMI, waist circumference, physical activity measures, sedentary minutes, glycohemoglobin, HDL and total cholesterol, hs-CRP, uric acid, ALT, GGT, and systolic and diastolic blood pressure. The features were standardized using *StandardScaler* before clustering so that the contribution of these features was not dependent on scale of the units. These features were selected primarily as a result of literature review of previously done studies and understanding important cardiometabolic markers. The selected features are several of the risk factors used for diagnosing MetS in addition to MEC variables from NHANES which take physical measurements and lab data. In addition to this both age and sex were included as demographic variables, selected as a result of literature review.

For k-means the hyperparameters we selected were the number of clusters (k) and the initialization procedure. The k-means clustering was initialized using k-means++ centroid initialization (*init="k-means++"*) for convergence stability with 20 random initializations (*n_init=20*). This means that 20 initializations were done and the one with the lowest within-cluster inertia was selected. The default *n_init* with k-means++ is 1 but more runs are recommended for sparse or high-dimensional applications like ours. The choice of k , as discussed later, was evaluated using silhouette score and cluster stability across many runs. A *random_state=42* was also used for reproducibility.

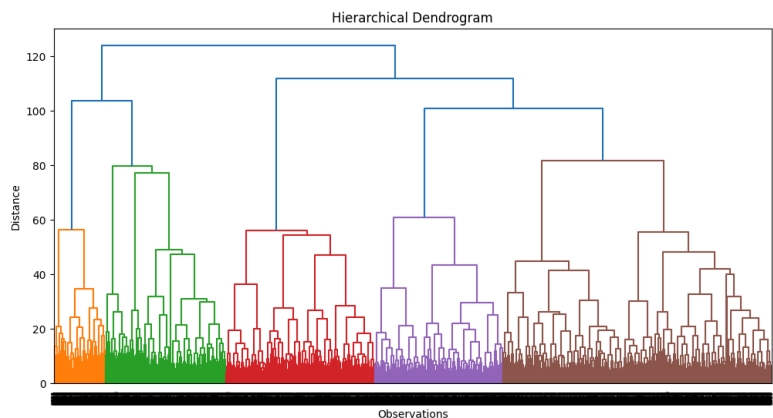
Agglomerative hierarchical clustering does not need random initialization like k-means and uses a hyperparameter of linkage criterion. This hierarchical clustering was initialized using Ward's linkage which was chosen because the NHANES data is diverse and noisy in its health measurements and Ward's method is the best at handling noise as stated in the documentation. Ward's linkage also minimizes the within-cluster variance which makes it more comparable to k-means while it still results in hierarchical structure. We did not do grid search for hyperparameter selection but chose our hyperparameters based on domain suitability and knowledge of NHANES, stability across runs, and clustering quality metrics.

Three metrics were used to evaluate the resulting clusters from the best model using k-means and the best model using agglomerative hierarchical clustering. The elbow method and silhouette score were both used in determining the optimal number of clusters, k , testing values between 3-10. The elbow method minimizes intra-cluster distance while the silhouette score measures how similar a data point is to its own cluster compared to other clusters, both producing metrics to measure the quality of the clusters produced. The Adjusted Rand Index (ARI) was also used for stability analysis of the clusters to measure if similar clusters were produced between the methods (k-means and hierarchical) and if similar clusters were produced by the same method if a subset of the data was used in the clustering process. This is important because data stability testing can tell us if the groupings we find are reflecting underlying patterns or just noise.

K=4 was the optimal number of clusters for k-means when evaluating values between 3-10. This range was selected for our analysis because fewer than three clusters would not capture enough differences to identify meaningful cardiometabolic health clusters, and more than ten clusters would likely create small and less interpretable groups. The optimal k was selected using a combination of silhouette score, elbow plot evaluation, and data stability. The elbow plot had no sharp bend and had very gradual flattening. Silhouette scores were uniformly low, near zero, across the tested range (0.08-0.12). Silhouette scores range from -1 to 1 and scores near zero indicate lots of points are either overlapping or sitting on or near a decision boundary, in this case likely because of overlapping of cardiometabolic phenotypes. K=4 had much higher stability under resampling as compared to higher k values. Combining these metrics, k=4 was determined to be the optimal k.

Figure 5

For Agglomerative Hierarchical clustering, K=5 was selected for the optimal number of clusters in the hierarchical model based on the dendrogram in figure 5. The dendrogram was examined for the largest gap in merge distance between partitions, a standard way for selecting a natural cluster cut point (Everitt et al., 2011). The largest gap between merges appears between a distance of approximately 80 and 100 as visualized in Figure 5. Because of this the cut point was set to *color_threshold=90* in our dendrogram and the corresponding best k is 5. These 5 resulting groups are the groups represented in orange, green, red, purple, and brown below in the dendrogram in Figure 5. Silhouette analysis was also done using k values from 3-10. Ward's linkage was used and silhouette scores were low (<0.15) across all k-values. Because these scores were consistently low, the dendrogram was ultimately used to determine the hierarchical model would use k=5.



In order to compare the best model from k-means and hierarchical clustering we compared data stability and also took into account silhouette score. K-means and hierarchical clustering were both evaluated on data stability using 100 iterations run on 80% random subsampling. Labels were then compared with ARI to measure similarity between the original labels and the cluster labels of each subset. K-means had high data stability mean ARI: 0.907 ± 0.034 which shows that the cluster assignments were highly similar for subjects even if the clusters were formed on a subset of the data. Hierarchical clustering had a much lower data stability score with mean ARI: 0.241 ± 0.074 . This shows that the hierarchical solution is sensitive to which subjects are included in the clustering, which is a limitation for generalizability of the hierarchical method.

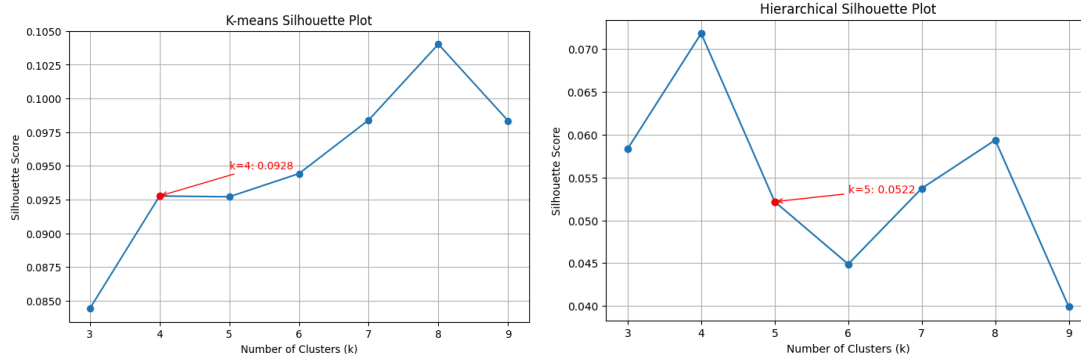
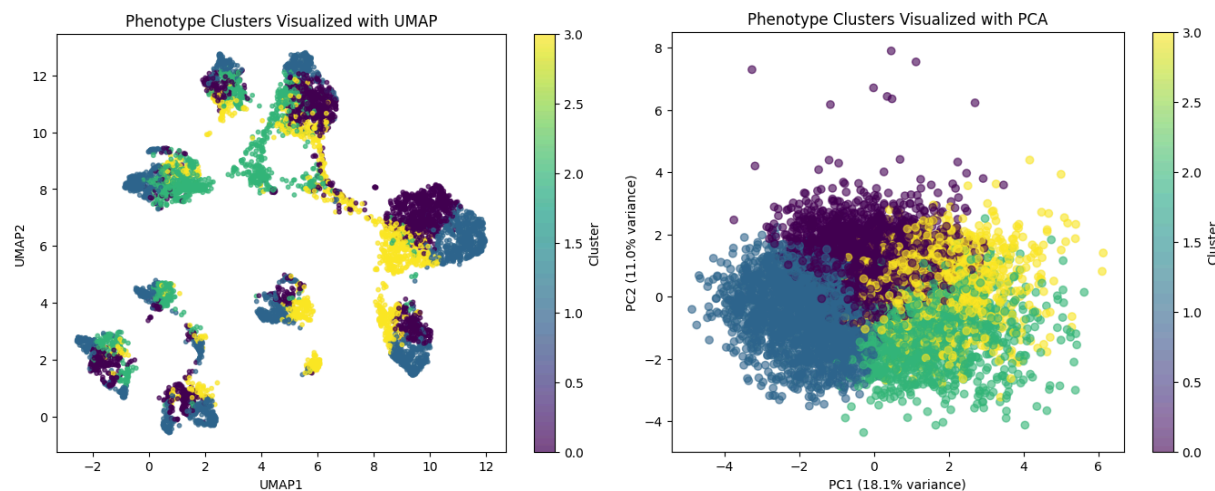


Figure 6

K-means (k=4) was selected as the final model to create the cardiometabolic phenotypes. While both methods had low silhouette scores, k-means marginally outperformed hierarchical clustering on silhouette score (0.0928 vs 0.0522). K-means produced significantly more stable clusters when tested for data stability (mean ARI 0.907 vs 0.241) visualized in Figure 6. Sensitivity analysis was also done on the k-means solution to test initialization stability. This was done by comparing cluster membership across two models with different random_state initializations, resulting in mean ARI 0.972 ± 0.031 . This high score shows the model is stable and reproduces the same clusters even if initialization points change. High ARI scores for both data stability and initialization stability alongside a low silhouette score suggests that the clusters do find reproducible and stable groups but that these groups are overlapping or irregularly shaped.

To visualize how well these phenotypes were separated we projected the full set of standardized cardiometabolic and demographic variables into two dimensions using UMAP (Uniform Manifold Approximation and projection), pictured in Figure 7 (left). The visually distinct groups in the UMAP plot tend to contain subjects of several colors (phenotypes). While the four clusters may capture broad clinical differences between groups, this shows the clusters may not capture further subgroup structure. We also visualized the same data using principal component analysis (PCA) shown in Figure 7 (right). PCA reduces the data to the two directions that explain the most variance in the data, and this plot shows more overlap between the groups but the four clusters have somewhat defined areas on this visualization. Both of these visualization methods show overlap between phenotypes suggesting that for these subjects, cardiometabolic risk is more of a spectrum of overlapping health characteristics and not smaller distinct groups.

Figure 7



While the reduced dimensionality plots did not show visually distinct groups we are still able to analyze the four phenotypes produced as a result of our best k-means model. The clustering variables vary in scale so before summarizing the phenotypes the variables were standardized. Without standardization the variables with large numeric values, for example sedentary minutes, would dominate the color scale. In order to standardize the features we converted each of the features in the cluster summary to a z-score by subtraction the mean of the variable and dividing by the standard deviation. This puts all the variables on the same scale so that when reading the heatmap the colors can be interpreted as relative differences between the clusters. When interpreting the heatmap below, positive values (red) indicate that a cluster is above the sample average for that feature, and negative values (blue) show the cluster is below average in that feature.

One other important note in interpreting the heatmap is that the *Moderate Work Activity* and *Vigorous Work Activity* and *Walk or Bicycle for Transportation* features are surveyed by asking a question similar to *Does your work involve moderate (or vigorous)-intensity activity?* And *do you walk or use a bicycle to get to and from places?* The

responses are encoded with values of 1: yes, and 2: no. This means that for this specific feature, a higher number (and deeper shade of red) actually represent less moderate/vigorous activity at work, and lower numbers (deeper blue) represent higher amounts of activity at work, and higher values indicate less walking and biking while lower values represent more.

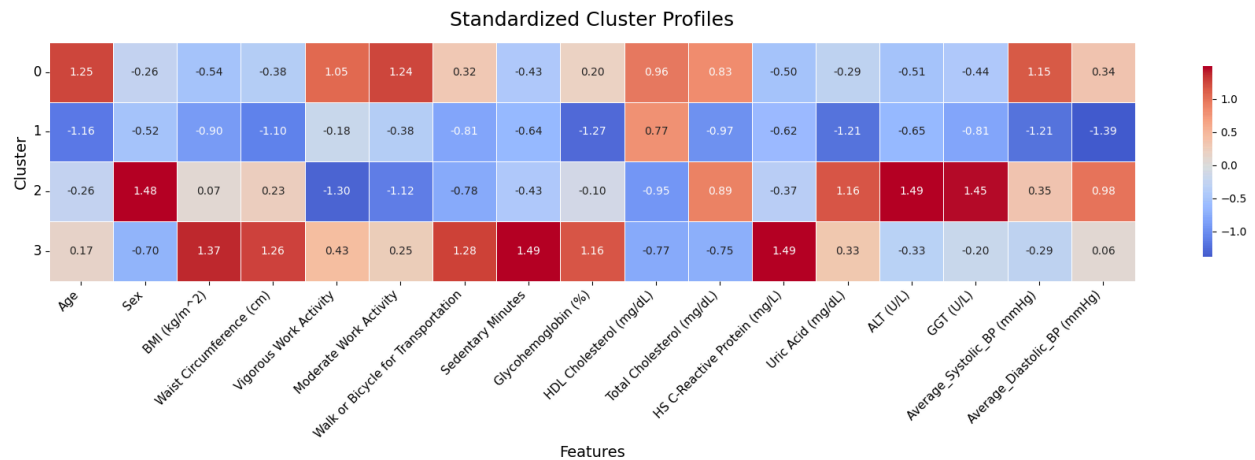


Figure 8

From the standardized cluster heatmap in Figure 8 and average feature value table in Table 5 we are able to describe the four phenotypes based on the results. **Phenotype 0 (Age / BP driven risk)**: This group has the highest average age at about 64.6 years. The average systolic BP is the highest among groups but the group has a moderate average diastolic BP. This group has relatively low sedentary minutes and the least amount of vigorous and moderate work activity. **Phenotype 1 (Good cardiometabolic health, young)**: This is the youngest group with an average age of 35.0 years. This group also has the lowest BMI and waist circumference on average. The average blood pressures are healthy and this group has the lowest average glycohemoglobin, HS-CRP, and total cholesterol. **Phenotype 2 (Metabolic/ Liver Risk)**: Middle aged with average of 46.0, 90% male, BMI 31.4 (Class 1 obesity), waist circumference 107 cm. ALT and GGT are both the highest on average in this phenotype, suggesting hepatic involvement. This group also has the highest amount of moderate and vigorous work intensity across groups (survey answers encoded 1 for yes, work involved moderate/vigorous work and 2 for no). **Phenotype 3 (Obese, cardiometabolic risk)**: This group is late-middle aged with an

Cluster Feature Averages by Phenotype				
Phenotype	0	1	2	3
Age	64.58	35.04	46.02	51.35
Male percentage	0.45	0.38	0.90	0.33
BMI (kg/m^2)	27.57	25.30	31.41	39.54
Waist Circumference (cm)	97.65	86.88	106.79	122.25
Vigorous Work Activity	1.91	1.70	1.50	1.80
Moderate Work Activity	1.73	1.48	1.37	1.58
Walk or Bicycle for Transportation	1.80	1.71	1.71	1.87
Sedentary Minutes	325.08	317.09	325.25	399.14
Glycohemoglobin (%)	5.96	5.33	5.83	6.37
HDL Cholesterol (mg/dL)	58.19	56.92	45.35	46.52
Total Cholesterol (mg/dL)	194.59	172.90	195.28	175.55
HS C-Reactive Protein (mg/L)	2.59	2.19	3.02	8.99
Uric Acid (mg/dL)	5.30	4.61	6.40	5.77
ALT (U/L)	17.68	16.26	38.44	19.50
GGT (U/L)	24.49	17.89	57.95	28.62
Average_Systolic_BP (mmHg)	137.02	110.23	127.95	120.71
Average_Diastolic_BP (mmHg)	76.78	67.47	80.20	75.27

Table 5

average age of 51.3 years. This group has the highest average BMI and sedentary minutes, and lowest work activity. This group has an extremely high HS-CRP of around 8.99 while other groups all have levels around 2-3. The glycohemoglobin is also significantly higher than the other groups at 6.37 on average, close to the levels of those with type 2 diabetes which is 6.5%.

Discussion:

The supervised learning analysis delineated that available demographic, behavioral, and cardiometabolic variables do contain important information for predicting reported sleep disorder, but that information is only moderately strong and is driven mostly by broad risk patterns rather than highly specific sleep-related markers. What was most surprising was that Logistic Regression performed slightly better than the more complex tree-based models. We believe that the predictive pattern in our data was more linear and general than we initially expected. We also found that adding the phenotype label did not improve performance. This was especially important as it showed that the phenotypes were more useful for interpreting participant groupings than for adding new predictive value.

One challenge was balancing model performance in an imbalanced classification setting. We found that accuracy alone would have overstated performance while hiding missed sleep disorder cases. We addressed this by tuning models with F1 score, evaluating multiple metrics, and using threshold analysis to better understand the precision-recall tradeoff. Another challenge was that the model often seemed to rely on general cardiometabolic risk as a proxy for sleep disorder. This often led to false positives for high-risk participants and false negatives for younger and healthier participants. Possible work or future endeavors could include utilizing more direct sleep related variables, such as sleep duration, insomnia symptoms, stress, mental health, medication use, or work schedule. This may help the model capture sleep disorder cases that are not well explained by metabolic risk alone.

The unsupervised learning portion of this project allowed us to learn a lot about the structure of the NHANES data. One of the most interesting learnings was that the clusters that resulted from k-means could perform well in our data stability tests while not performing well on the silhouette score. This was a learning experience because lots of clustering problems that we learn from have strong resulting clusters when the optimal k is chosen meaning they have data stability and they have high silhouette scores. This allowed us to explore in-depth imperfect clusters.

It was surprising that despite near-zero silhouette scores across the board and a steadily decreasing elbow plot with no sharp bend, the resulting clusters were still relatively distinct in average value which we can see in both the heatmap and in the summary table. This was surprising because near-zero silhouette scores typically indicate that lots of the points are on the boundaries between two clusters and that clusters likely overlap. Because of this it was surprising that the cluster summaries showed lots of features with different average values.

There were lots of challenges in the unsupervised section. One of the first problems encountered was the large amount of missing values in the NHANES data which need to be dealt with prior to clustering. We addressed this by evaluating the missingness and understanding why certain variables had high missingness, for example, missingness for insulin, triglycerides, and fasting glucose is mostly due to NHANES study design. These biomarkers require fasting, so NHANES only collects them from a smaller fasting subsample rather than the full MEC exam sample. Because of this, we exclude those variables with high missingness so that when dropping null values from our dataframe we still retain subjects without those values. It was also a challenge to ensure we were selecting appropriate variables in this unsupervised section beyond just based on missingness. For this we primarily relied on literature review for selection of the cardiometabolic features as well as demographic features that would be included.

With more time we would potentially extend the unsupervised section to cluster on different variables beyond just those derived from literature review. We would select subsets potentially based on quantitative and exploratory analysis. We would also be interested in clustering strictly on lifestyle variables and evaluating those resulting clusters and seeing if those phenotypes improve the combined model.

Ethical Considerations:

Our project uses public, de-identified NHANES data, which reduces privacy risks because individual participants cannot be directly identified. Results should be reported at the group level and not used to make clinical judgments about any individual person. Another ethical concern is bias and generalizability. NHANES is designed to represent the U.S. population, but our machine learning models use a complete-case analytic sample and are not fully survey-weighted, so results may not generalize equally to all groups. Because the data are cross-sectional, our models can identify associations and prediction patterns, but they should not be interpreted as showing that cardiometabolic factors cause sleep trouble. Finally, any real-world use of this type of model would require caution because false negatives could miss people with sleep problems, while false positives could incorrectly label people as at risk. For this reason, the model should be viewed as an exploratory screening and research tool rather than a diagnostic system.

Statement of Work:

Yasthil Singh contributed to dataset acquisition, NHANES module merging, data cleaning, feature engineering, and creation of the main analytic dataframes, including `metS_df` and `cardio_met`. He also contributed to the literature review, data source documentation, feature list, ethical considerations, and report editing.

Samantha Hastie contributed to the unsupervised learning section, including k-means clustering, agglomerative hierarchical clustering, cluster evaluation, stability analysis, and phenotype interpretation. She also created and interpreted the unsupervised visualizations, including silhouette plots, PCA/UMAP plots, and the standardized phenotype heatmap.

Seema Guruvadoo contributed to the supervised learning section, including model development, evaluation of supervised models, comparison of baseline and phenotype-enhanced models, model interpretation, sensitivity analysis, and failure analysis. She also contributed to supervised evaluation outputs such as performance metrics, confusion matrices, feature importance, ablation analysis, sensitivity analysis, and failure analysis.

References:

1. Everitt, B. S., Landau, S., Leese, M., & Stahl, D. (2011). *Cluster analysis* (5th ed.). John Wiley & Sons.
2. Huang, A. A., & Huang, S. Y. (2023). Use of machine learning to identify risk factors for insomnia. *PLoS ONE*, 18(4), e0282622. <https://doi.org/10.1371/journal.pone.0282622>
3. Huang, W., Ramsey, K. M., Marcheva, B., & Bass, J. (2011). Circadian rhythms, sleep, and metabolism. *Journal of Clinical Investigation*, 121(6), 2133–2141. <https://doi.org/10.1172/JCI46043>
4. Kassi, E., Pervanidou, P., Kaltsas, G., & Chrousos, G. P. (2011). Metabolic syndrome: Definitions and controversies. *BMC Medicine*, 9, 48. <https://doi.org/10.1186/1741-7015-9-48>
5. Smiley, A., King, D., & Bidulescu, A. (2019). The association between sleep duration and metabolic syndrome: The NHANES 2013/2014. *Nutrients*, 11(11), 2582. <https://doi.org/10.3390/nu11112582>
6. Sabbagh, M. I. R., Oliveira, T. de, Souza, D. R. de O., Avelar, M. L. D., Silva, E. da, Santos, G. A., Lopes, G. F., & [remaining authors if listed in your source]. (2025). Social jetlag, sleep, and metabolic syndrome in adults: Insights of circadian misalignment from NHANES 2017–2020. *Sleep Science and Practice*, 9, Article 58. <https://doi.org/10.1186/s41606-025-00158-3>

Appendix:

Final Supervised Learning Predictors	Fasting Subsample Sensitivity Features in metS_df
Age	Waist Circumference (cm)
Sex	Average_Systolic_BP (mmHg)
BMI (kg/m^2)	Average_Diastolic_BP (mmHg)
Waist Circumference (cm)	HDL Cholesterol (mg/dL)
Vigorous Work Activity	Fasting Glucose (mg/dL)
Moderate Work Activity	Triglycerides (mg/dL)
Walk or Bicycle for Transportation	Fasting Weight
Sedentary Minutes	
Glycohemoglobin (%)	
HDL Cholesterol (mg/dL)	
Total Cholesterol (mg/dL)	
HS C-Reactive Protein (mg/L)	
Uric Acid (mg/dL)	
ALT (U/L)	
GGT (U/L)	
Average_Systolic_BP (mmHg)	
Average_Diastolic_BP (mmHg)	